Curriculum Vitae

Anjali Rana | PhD Student – Senior Research Fellow | Department of Biological Sciences | Indian Institute of Science Education and Research (IISER), Mohali | India | 140306 | phdwithanjali@gmail.com

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PUBLICATIONS

Rana, A., Verma, P., Pandit, S. B., Werren, J. H., Wang, X., & Raychoudhury, R. (2025). Genome assembly and annotation of the parasitoid jewel wasp *Nasonia oneida*. ***Scientific Data***, 12(1), 1096.

Nain, D., *__Rana, A.__, Raychoudhury, R., & Sen, R. (2025). Parasite-Induced Replacement of Host Microbiota: Impact of *Xenos gadagkari* Parasitization on the Microbiota of *Polistes wattii*. ***Microbial Ecology***, 88(1), 20.
*__First Author__ – Equal Contribution

Panchal, A., Sen, R., Agarwal, R., **Rana, A.**, & Raychoudhury, R. (2025). Fungus-farming termites can protect their crop by confining weeds with fungistatic soil boluses. ***Science***, 389(6767), 1366-1371.

Nain D, **Rana A**, Raychoudhury R, Sen R. Morphology, biology and phylogeny of *Xenos gadagkari* sp.nov. (Strepsiptera: Xenidae): an endoparasite of *Polistes wattii* (Hymenoptera: Vespidae). ***Zootaxa***. 2024;5493(5):561-576.

Under Review:

Microbiota of polistine wasps: a comparison of four species from India and beyond
Rana, A., Nain, D., Raychoudhury, R., & Sen, R

Under Preparation:

Host-symbiont-microbe interactions shape the microbiota of the jewel wasp, *Nasonia*
Rana, A., and Raychoudhury, R.

EDUCATION

- **PhD:** Evolutionary Genetics - Department of Biological Sciences, IISER Mohali, Punjab, India (*Thesis submitted in October 2025*)

- **Title of Thesis**: Disentangling a Holobiont: Role of microbes in the ecology and evolution of the jewel wasp, *Nasonia*

Date of Admission - 05/08/2019
Expected Year of Award - February 2026

RESEARCH WORK

During my PhD, I studied the role of various microbes, including bacteria, fungi, and viruses, in the ecology and evolution of insects during my PhD. I have investigated the different factors that contribute to the microbiome (bacterial community), mycobiome (fungal community), and virome (viral community) of the jewel wasp, *Nasonia*. Therefore, I first characterized the different microbial communities of these insects and then studied host-symbiont-microbiome interactions. Additionally, I assembled the genome of one of the youngest *Nasonia* species, *N. oneida,* and performed comparative genomics with other *Nasonia* species and other hymenopteran insects.

I have also worked with social insects, including highly eusocial and primitively eusocial species, such as termites (*Odontotermes obesus*) and paper wasps (*Polistes* spp.), which have shaped my perspective on host biology across different levels of social organization.

SKILLS

Molecular Techniques including DNA Isolation, RNA Isolation, Polymerase Chain Reaction (PCR), quantitative PCR (qPCR), NGS library preparation and sequencing (Illumina and Nanopore), Data Analysis (Genome, Microbiome and Transcriptome), Bioinformatics, and Microbial culturing.

Reviewer

Reviewer for peer-reviewed scientific journals, including Animal Microbiome, Microbial Ecology, and Discover Developmental Biology.

FELLOWSHIPS, GRANTS, HONORS

- ANRF International Travel Grant Award to attend the *ESEB2025* 2025
- Best Poster Award (*26th Punjab Science Congress*) 2023
- Best Poster Award (*International Conference on Antimicrobial Resistance and Microbiome*) 2022
- CSIR NET JRF- All India Rank- 56 2018

ORAL PRESENTATIONS

- "*Conducting Experiments in Microbial Ecology: Identifying Microbes in Insects Using Nanopore Sequencing*" Workshop SGGS College 2025, 2nd February, SGGS College, Panjab University, Punjab, India

- "*Genome Assembly and Annotation of Nasonia oneida: A valuable addition to the Nasonia Model System*" *Nasonia* Meeting 2023, July 28-29, Muenster, Germany.

- "*Impact of Endosymbiont and Host Genotype on the Microbiome of Nasonia*" DBS Seminar 2024, 6th January, Department of Biological Sciences, IISER Mohali, Punjab, India.

POSTER PRESENTATIONS

- "*Genome assembly and annotation of the parasitoid jewel wasp Nasonia oneida*, with insights into the genomic evolution of *Nasonia* species" at **ESEB2025**, August 17-22, **Barcelona, Spain**.

- "*Wolbachia, an insect reproductive manipulator, can impact the microbial community of Nasonia vitripennis*". **26th Punjab Science Congress**, 7-9 February 2023, SGGSW University, Fatehgarh Sahib, **Punjab, India**.

- "*Wolbachia - an insect reproductive manipulator, alters the microbial community of Nasonia vitripennis*." **International Conference on Antimicrobial Resistance and Microbiome under Changing Climate**,10-12 October 2022, Pondicherry University, **Puducherry, India,** in association with **Imperial College, London**.

- "*Role of Viruses in the Genome Evolution of Nasonia*" **Indian Society of Evolutionary Biologists**, 16-18th February 2022, Punjabi University, Patiala, **Punjab, India**.

SCIENTIFIC MEMBERSHIP

- British Ecological Society (BES)
- American Society of Microbiology (ASM)
- European Society for Evolutionary Biology (ESEB)
- Society for the Study of Evolution (SSE)
- The Association for the Study of Animal Behaviour (ASAB)